

Category	BIOS	UEFI
Definition	Traditional firmware interface used to initialize hardware and start the boot process.	Modern firmware interface designed to replace traditional BIOS.
Boot Process	Performs hardware initialization and searches for a boot device.	Initializes hardware and uses a modern boot manager to locate and start the operating system.
Maximum Disk Support	Traditional BIOS systems commonly use MBR, which has limitations on large disks.	Commonly works with GPT, allowing support for much larger disks.
Partition Style	Commonly associated with MBR.	Commonly associated with GPT.
Security Features	Limited compared with modern firmware.	Provides modern security capabilities such as Secure Boot.
Speed	Generally has a simpler and older startup process.	Provides a more modern and efficient boot process.
Advantages	Simple, widely supported on older systems, and familiar.	Modern architecture, large-disk support, GPT compatibility, and stronger security features.
Disadvantages	Older design, limited partition/disk capabilities, and fewer modern security features.	More complex than traditional BIOS and may require compatible hardware and operating-system support.
Modern Usage	Mainly found on older computers and legacy systems.	Widely used on modern computers.

UEFI has largely replaced traditional BIOS because it provides a more modern firmware environment for current computer systems. One important advantage is its support for GPT partitioning, which is better suited to modern storage devices and large-capacity disks. UEFI also provides improved security capabilities, including Secure Boot, which can help prevent unauthorized or untrusted boot software from loading during startup.

Another advantage of UEFI is its modern boot-management architecture. Instead of relying entirely on the traditional BIOS boot process, UEFI can directly work with modern bootloaders and operating systems. This makes it better suited to current operating systems and hardware.

BIOS remains important for understanding older computers and legacy systems, but UEFI is generally preferred for modern systems because of its improved capabilities, compatibility with modern storage technologies, and security features.